

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of the following could be used as a test for autocorrelation up to third order?' is 'The Breusch-Godfrey test'.
2. True or False: The correct answer to 'What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$ ' is '0'.
3. True or False: The correct answer to 'An example of combining two substances is' is 'putting clothes in a suitcase'.
4. True or False: The correct answer to 'when the needle of a compass lines up with Earth's magnetic poles , the needle is' is 'pointing west'.
5. True or False: The correct answer to 'Which best describes the orbit of the Moon around Earth?' is 'elliptical with a period of about one month'.
6. True or False: The correct answer to 'When a vegetative organism lifeform goes into a state of creating a life from itself, what is it called?' is 'seed harvest'.
7. True or False: The correct answer to 'Cerebral spinal fluid is secreted by the _____' is 'Pia mater'.
8. True or False: The correct answer to 'A student waiting for class to be over, in the last minute until the bell rings' is 'counts down portions'.
9. True or False: The correct answer to 'Pumice is formed when lava from a volcano cools. Which rock type is pumice?' is 'Sedimentary rock'.
10. True or False: The correct answer to 'The xylem absorbs water which end of a plant' is 'left side'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to split a list for every nth element.
2. Write a python function to find the difference between largest and smallest value in a given array.

End of Paper